

CHEM1141	ROLE OF CHEMISTRY IN DAILY LIFE	L	T	P	S	J	C
		2	0	0	0	0	2
Pre-requisite	None						
Co-requisite	None						
Preferable Exposure	None						

Course Description:

This course explores the fundamental principles of chemistry and their practical applications in everyday experiences. Students will gain an understanding of how chemical concepts influence the food we eat, the medicines we use, the cleaning products in our homes, and even the air we breathe. Common materials like disinfectants, pesticides, and cleaners, and their emphasis is placed on making informed, science-based decisions in daily life while appreciating the impact of chemistry on health, safety, and the environment. Through interactive discussions and real-life examples, the course bridges the gap between theoretical chemistry and its real-world relevance.

Course Educational Objectives:

- To gain a clear understanding of fundamental chemistry terms and concepts such as atoms, molecules, elements, compounds, and mixtures
- To understand how chemical principles and substances are involved in and influence the products, processes, and phenomena we encounter in our daily lives
- To analyze and understand the chemical composition and properties of water to ensure its quality and , safety
- To understand about disinfectants, pesticides, and cleaners are and how they are used in daily life
- To identify the composition and quantify the amounts of substances present in a sample

UNIT 1 BEGINNER'S CHEMISTRY

6 Hrs

What is Chemistry?, Relationship between Chemistry and other branches of Science, Branches of Chemistry, Examples of Chemistry in our daily lives- Importance of Chemistry in food

UNIT 2 CHEMISTRY IN EVERYDAY LIFE

6 Hrs

Chemicals in food- Chemical structure, properties, source, outlines of the action of any two of preservatives, artificial sweetening agents, antioxidants, Cleansing agents- soaps and detergents, cleansing action, Chemicals in medicines- Chemical structure, properties, source, outlines of the action of any two each of analgesics, tranquilizers, antiseptics, antibiotics, antacids

UNIT 3 WATER CHEMISTRY

6 Hrs

Types of water- types of impurities in water-Physical and chemical properties, Water pollution- Major sources of water pollution, Properties of pure water- Factors affecting natural water, Water quality criteria: Water quality parameters for wastewater, industrial water, and domestic water.

UNIT 4 ELEMENTARY IDEA OF DISINFECTANTS, PESTICIDES, AND CLEANERS

6 Hrs

Chemical structure, properties, sources ; Alcohol based hand sanitizers, sodium hypochlorite, , naphthalene, antiseptic solutions, Pesticides and insecticides like DDT, Mosquito repellent, Boric acid, Toilet cleaners, Domestic phenols, and Floor cleaners

UNIT 5 INTRODUCTION TO CHEMICAL ANALYSIS

6 Hrs

Introduction - Types of chemical analysis-purpose of chemical analysis, wet chemical Methods-Titrations, Color-forming reactions-precipitations, Basic analytical instruments: pH, Colorimeter, and Conductometer

Textbook(s):

1. Galen S. Ewing, *Instrumental methods of Analysis*, 5th ed., McGraw Hill Higher Education, 1985.
2. Claude E. Boyd, *Water quality in ponds for aquaculture*, Alabama Agricultural Experiment Station, 1990.
3. Singh, Kirpal, *Chemistry in daily life*, PHI Learning Pvt. Ltd., 2012.
4. Galen S. Ewing, *Instrumental methods of Analysis*, 5 th edition ed., McGraw Hill Higher Education, 1985.

Reference(s):

Course Outcomes:

1. By the end of the course, the student will be able to Explain the importance of Chemistry for different branches of Science and in daily life
2. Understand the importance of different chemicals in food and medicines
3. Able to describe the quality parameters of water
4. Will know the applications of chemistry in the daily life
5. Able to explain the principles underlying the chemical analysis

Course Articulation Matrix:

B.Sc. Biochemistry / B.Sc. Biotechnology / B.Sc. Chemistry / B.Sc. Electronics / B.Sc. Environmental Science / B.Sc. Food Science & Technology / B.Sc. Mathematics / B.Sc. Microbiology / B.Sc. Physics / B.Sc. Statistics

	POs																PSOs			
CO	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	1	2	3	4
1	2	2	1		2	2			2			1	1				2	2	2	
2	2	2	1		2	1			2			1	2		2		2	2	1	2
3	1	1	1		1	2			1			1	1		1		1	1	2	1
4	1					1	2	1	1		1	1	3	3	3	1	1	1	1	3
5	1	1	1		1	2			1			1	1		1		1	1	2	1

B.A. Economics / B.A. English / B.A. History / B.A. Political Science / B.A. Psychology / B.A. Sociology / B.A. with major in Cinematic Arts and Design / B.A. with major in Journalism and Media Studies

	POs																PSOs			
CO	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	1	2	3	4
1	2	2	1		2	2			2			1	1				2	2	2	
2	2	2	1		2	1			2			1	2		2		2	2	1	2
3	1	1	1		1	2			1			1	1		1		1	1	2	1
4	1					1	2	1	1		1	1	3	3	3	1	1	1	1	3
5	1	1	1		1	2			1			1	1		1		1	1	2	1

B.Com. (ACCA) / Bachelor of Business Administration / BBA (Business Analytics) / BBA (Financial Markets)

	POs					PSOs			
CO	1	2	3	4	5	1	2	3	4
1	2	2	1	1	2	2	2	1	
2	2	2	2	1	2	2	2	2	1
3	1	2	1	1	2	1	1	2	2
4	1	2	2	2	3	2	2	3	3
5	2	2	2	1	2	2	2	2	3

3 – High, 2 – Medium & 1 – Low Correlation

APPROVED IN MEETINGS HELD ON:

BOS : 23-09-2025

Academic Council Number: 35

Academic Council : 24-10-2025

SDG No(s). & Statement(s) :

6 & Clean Water and Sanitation : Ensure availability and sustainable management of water and sanitation for all.

SDG Justification(s):

Access to safe water, sanitation and hygiene is the most basic human need for health and well-being. Demand for water is rising owing to rapid population growth, urbanization and increasing water needs from agriculture, industry, and energy sectors.